

Single LLMs vs. Autonomous Agents

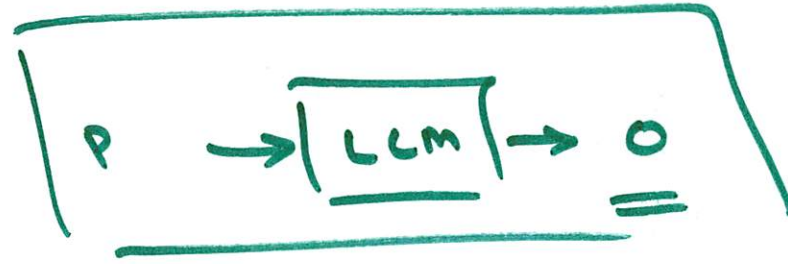
- Single LLM Evaluation

- Static

- Stateless

- Single inference

- Output-focused



- Agent Evaluation

- Stateful

- Multi-step

- Tool-dependent

- Workflow-focused

Phase

Early Dev

Late Dev

Prod

High Risk Prod

Mature low risk Prod

Likely Mode

HITL

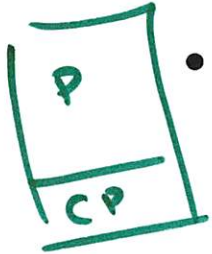
HITL → HOTL Int

Mostly HOTL

HITL

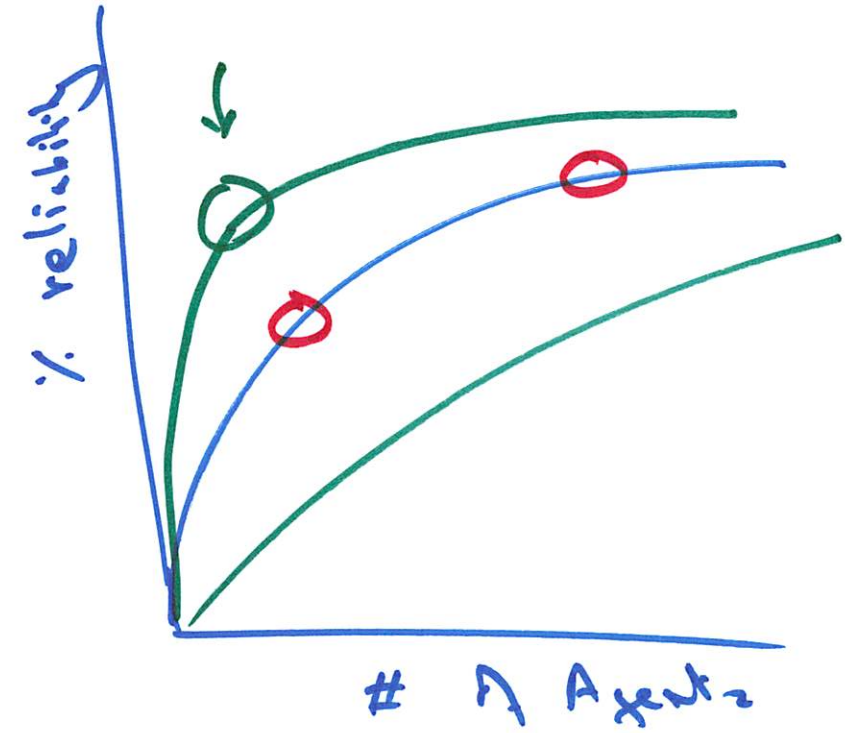
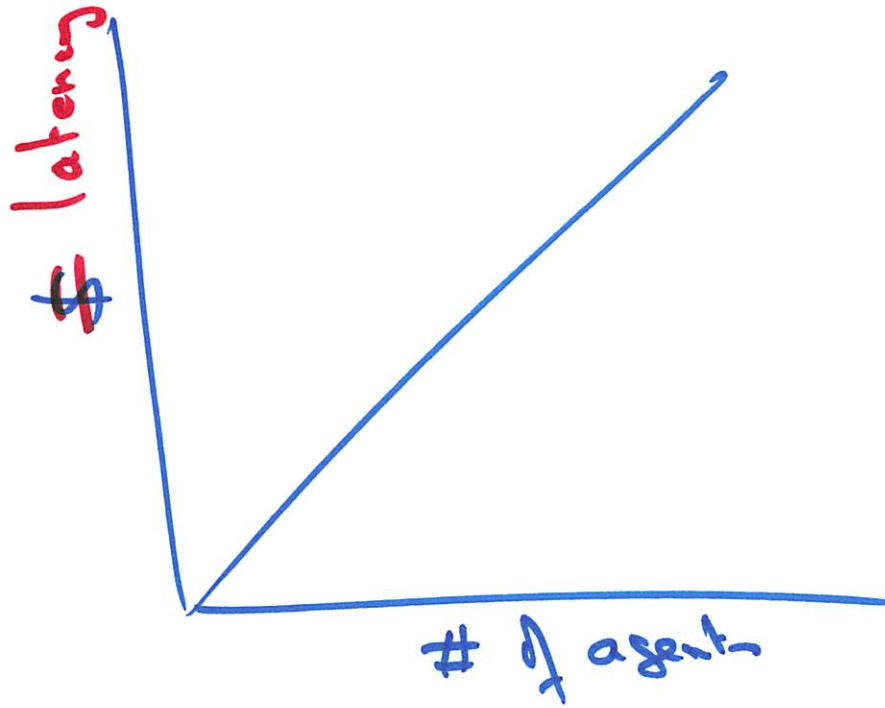
Selective HOTL

Establishing Output Confidence



- Self-Reflection (Critic): Using a "Critic" prompt to ask the model if it is confident that the output is correct relative to the input.
- The Critic is asked: "Are you confident about the argument that you made? Is it logical?"
- Rubric-Based Scoring (Human criteria): Evaluating the output against a human-provided rubric (e.g., check for source usage, formatting, and constraint adherence).
 - Did the CON agent cite two verified sources? Did it avoid ad hominem attacks?
- Consensus-Based Confidence (Wisdom of the crowd) by having the agent generate 5–10 different reasoning paths at a higher temperature. If the majority of paths arrive at the same conclusion, the confidence is high.
- A panel of 5 Judge agents evaluates the debate. Majority dictate the winner





Sensitivity in Agent Systems

- Axis 1 — World Coupling
 - Informational Systems
 - Read inputs
 - Produce advice or analysis
 - No direct world modification
 - Actuating Systems
 - Write to external systems
 - Trigger transactions or actions
 - Directly modify digital or physical environments
- Axis 2 — Irreversibility of Impact
 - Reversible (low cost to undo)
 - Costly to reverse
 - Irreversible / catastrophic

